

4 Input files

There are three files describing the bins, the items, and the parameters. The files are csv files: values on a line are separated by semicolons (;).

4.1 The file `input_bins.csv`

The file `input_bins.csv` contains m lines, each containing the description of one bin. A line contains the bin type (a string), the length, width, height and maximum weight allowed in the bin.

The following line corresponds with bin type CT40H of dimensions (12090, 2350, 2680) and maximum allowed weight 26400.

```
CT40H ; 12090 ; 2350 ; 2680 ; 26400
```

4.2 The file `input_items.csv`

The second file contains n lines, each containing the description of an item.

Each line contains the item identifier, its length, width, height, weight, the package material, the orientation constraint and the identifier of the actual product it contains.

For example, the following line describes item 0, of size (1200, 1000, 745), of weight 60.900, made of cardboard, and corresponding with product id 165559019RG. A blank field for the orientation constraint means that the orientation is free.

```
0 ; 1200 ; 1000 ; 745 ; 60.900 ; CARDBOARD ; ; 165559019RG
```

We now give another example, related to item 30 with a fixed orientation.

```
30 ; 1740 ; 1140 ; 850 ; 406.732 ; CARDBOARD ; LENGTHWISE ; 112845638RG
```

4.3 The file `parameters.csv`

The third file contains the list of parameter values, one per line. For example, if $\gamma = 0.9$, $\alpha = 2000$, $\beta^L = 5$, $\beta^R = 10$, $\delta^S = 0.05$, $\delta^L = 0.05$, $\delta^R = 0.05$, and $\mu = 750$, the file will be the following.

```
tolerance_item_product_demands=0.9
maximum_density_allowed_kg=2000
max_number_of_rows_in_a_layer=5
max_number_of_items_in_a_row=10
layer_size_deviation_tolerance=0.05
row_size_deviation_tolerance=0.05
item_size_deviation_tolerance=0.05
maximum_weigth_above_item_kg=750
```

Note that the maximum density allowed is expressed in Kg/m², whereas the sizes are expressed in mm.

5 Output files description

A solution is described in five different csv files, each one corresponds with a level of the solution (bins, stacks, layers, rows, items). The values on each line are separated by semicolons.

For all files, except for the one dedicated to bins, the output data are similar. First, the unique identifier (numbered from 0) is given, then the identifier of the structure that contains the object, then the coordinates of two opposite corners. All coordinates are expressed with respect to the origin of the bin. The coordinates have to follow this convention. The origin corner is the closest to the origin corner of the bin. The extremity corner is the furthest one. The orientation of the items will be deduced from the coordinates.

All identifiers are unique, even if two layers (resp. rows) are in different bins (resp. layers), they cannot have the same id.

For readability reasons, the first line of each file describes the names of the fields.

5.1 The file bins.csv

For each bin, a line is written. Let $\hat{m}$ be the number of bins used. Each line contains a bin identifier (from 0 to $\hat{m} - 1$) and the corresponding bin type. For example, if bin 0 is of type CT40H, the line will be as follows.

```
0 ; CT40H
```

5.2 The file stacks.csv

There is one line per stack in this file. On each line, are given in this order: the identifier of the stack, the identifier of the bin that contains the stack, the origin and extremity coordinates. The value z_O is omitted, since it is always equal to 0. For example, if the stack that has been numbered 15 by the program is packed in bin 3 with ground position (30, 40) to (130, 80) and height 90, the line will be the following. This implicitly means that the stack has dimensions (100, 40, 90).

```
15 ; 3 ; 30 ; 40 ; 130 ; 80 ; 90
```

5.3 The file layers.csv

The layer file is similar to the stack file, except the z_O has to be specified. Each line contains the id of the stack in which the layer is packed, the origin and extremity coordinates, and one additional information. It specifies whether the layer is the top layer or not. This value will be 0 (not the top layer) or 1 (top layer). There can be only one top layer for each stack. For example, if the layer numbered 12, of size (99, 39, 35) is in stack numbered 15 at height 30 and is not the top layer, the line will be as follows.

```
12 ; 15 ; 30 ; 40 ; 30 ; 129 ; 79 ; 65 ; 0
```

5.4 The file rows.csv

The row file is the same as the stack file, except the embedding layer is given instead of the bin, and z_O has to be specified. For example, if row numbered 0, of size (30, 39, 35) is in layer 12 at position (60, 40, 30) the line will be the following.

```
0 ; 12 ; 60 ; 40 ; 30 ; 90 ; 79 ; 65
```

5.5 The file `items.csv`

The item file is similar to the row file. The identifiers of the items are specified in the input file. For example if item of id 3, of size (30,19,35) is in row 0 at absolute position (60,60,40), we obtain the following line.

```
3 ; 0 ; 60 ; 60 ; 40 ; 90 ; 79 ; 75
```